

ANNEE ACADEMIQUE: 2025/2026	LECTUERER : MALOBÉ LOTTIN CYRILLE M Tel : 677283965 / 620002732 Email : cyril.malobe@camtel.cm
CYCLE	DEGREE
SPECIALITE	NWS
NIVEAU	3
SEMESTRE	
INTITULE DU COURS	LINUX PROGRAMMING
VOLUME HORAIRE	

GENERAL OBJECTIVES

At the end of this course, students will be able to:

- 1- Use basic UNIX/Linux commands from the command line interface.
- 2- Organize and manage files within the Linux file system.
- 3- Manage and monitor processes in a Linux environment.
- 4- Combine UNIX tools using filters, pipes, redirection, and regular expressions.
- 5- Customize and manage the Linux working environment.
- 6- Write and execute Bash shell scripts for automation.
- 7- Use Linux programming tools to develop applications.

	TOPICS	SPECIFICS OBJECTIFS	PERIOD
01	Introduction to Linux and UNIX Environment	- Explain the evolution of UNIX and Linux systems - Differentiate between Linux kernel and GNU/Linux OS - Describe Linux architecture (kernel, shell, user space) - Distinguish between kernel space and user space - Identify major Linux distributions and their use cases - Navigate the Linux environment using the terminal	4
02	Linux Command Line and File System Management	- Execute essential Linux commands for file and directory management - Navigate the Linux file system efficiently using absolute and relative paths - Interpret and manage Linux file permissions	4

		(chmod, chown) - Analyze Linux directory structure and system organization - Monitor and manage running processes using system tools	
03	Filters, Pipes, Redirection and Regular Expressions	- Combine Linux commands using pipes to build processing chains - Use input/output redirection for file manipulation - Apply text-processing tools (grep, awk, sed) effectively - Construct and apply regular expressions for pattern matching - Analyze system logs and data using command-line utilities	4
04	Linux Shell Programming (Bash Scripting)	- Write and execute Bash scripts for automation tasks - Use variables, conditions, loops, and functions in scripts - Develop scripts that interact with files and system processes - Handle script inputs and arguments dynamically - Design automation solutions for system administration tasks	4
05	Linux Programming with C	- Write, compile, and execute C programs in Linux - Use standard I/O functions and libraries in C - Implement file handling operations in C programs - Debug programs using Linux tools (e.g., GDB) - Develop simple system-level utilities in C	4
06	Linux System Calls and Process Management	- Explain the role of system calls in Linux - Create and manage processes using fork(), exec(), and wait()	4

		- Handle process synchronization and lifecycle - Use signals for process control and communication - Analyze system behavior using process monitoring tools	
07	Linux System Calls and Process Management Recommend best security practices for enterprises	- Explain the role of system calls in Linux - Create and manage processes using fork(), exec(), and wait() - Handle process synchronization and lifecycle - Use signals for process control and communication - Analyze system behavior using process monitoring tools	4
08	Network Programming in Linux (Sockets)	- Explain client-server architecture in Linux - Implement basic socket programming (TCP/UDP) - Develop simple client-server applications - Use socket APIs (bind, listen, accept, connect, send, recv) - Analyze network communication behavior in Linux	4
09	GUI Programming Overview	- Describe the role of GUI frameworks in Linux - Differentiate between CLI and GUI applications - Identify major GUI toolkits (GTK, Qt) - Understand basic structure of graphical applications - Recognize use cases for GUI development in Linux	4

BOOKS / ARTICLES

- The Linux Programming Interface – Michael Kerrisk
- Advanced Programming in the UNIX Environment – W. Richard Stevens

- UNIX and Linux System Administration Handbook – Evi Nemeth
- Linux Command Line and Shell Scripting Bible – Richard Blum

Fait à Douala,

L'enseignant

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Le chef de département

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